

# Amey Ambade

## Senior Data Scientist | Product Lead, Machine Learning

AI Inventor | ML and Software Builder | Conference Chair | \$20M+ Impact | Advisor

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Open to roles in software development, ML engineering, Applied AI and Data Science.

## PROFESSIONAL SUMMARY

Senior AI / Lead ML Scientist with 7+ years delivering end-to-end, production ML software systems that generated \$20M+ measurable client value. Deep expertise across Generative AI (LLMs, RAG, SLMs), MLOps, and software development. Proven leader of cross-functional teams, strong consulting acumen, translating research into deployed products and complex ML into executive-level business outcomes.

## EDUCATION

### M.S. Computer Science (Machine Learning Specialization)

New York, NY | 2018

COLUMBIA UNIVERSITY

### B.Tech. Computer Science and Engineering

Indore, India | 2017

INDIAN INSTITUTE OF TECHNOLOGY (IIT)

## WORK EXPERIENCE

### SENIOR DATA SCIENTIST - MACHINE LEARNING | SLB (AI SOLUTIONS ORG) Houston, TX | 2023 – present

- Led cross-functional client teams delivering **LLM** based AI solutions for Fortune 500 energy clients, managing full project lifecycle from requirements gathering to production deployment, resulting in 15% operational efficiency gains across 50+ client sites, including deployment of finetuned SLMs for **IIoT Edge** applications.
- Implemented CompetitorIQ, an LLM-based document analysis and agentic framework using OpenAI APIs and **RAG** architectures, reducing client manual review time by 40% and enabling real-time insights from unstructured data about company competitors.
- Generated **\$3M (Q1 2024) SaaS revenue** delivering research-derived LLM applications integrated with internal DS orchestration platforms and external data lakes. Tech: **Python, Dataiku, Azure Foundry**.
- Built and productized the **SLB Vision Analytics suite** (YOLOv5, transfer learning, CNN) for live video: fire/smoke detection, line of fire, people counters, PPE detection, pose tracking — **\$4.5M revenue, 45% field safety improvement, 35% PPE risk reduction, and 30% surveillance time reduction**.
- Led a 3-person team to deliver an **end-to-end ML service** for dynacard **image classification** and real-time control optimization across **100+ wells**, resulting in **\$10M+ savings, 60% faster detection, and 19% downtime reduction**. Tech: **TensorFlow, Docker, Flask, IoT Edge**.
- Designed and deployed **Smart Surveillance** for hydrate detection (**\$2M+ revenue, +15% production, -12% labor**).

### DATA SCIENTIST | SCHLUMBERGER

Houston, TX | 2018 - 2023

- Developed **failure and anomaly prediction** models for Prognostic Health Management, implemented ML backend infrastructure and drove collaboration by partnering with field, client, and product teams to build production rate visualization with machine learning. Tech: **Azure ML, TensorFlow, Dataiku**
- Built a **deep learning**-based CNN model to classify erosion levels in mechanical tools and deployed a scalable cloud-based API, increasing precision by 8% and reducing detection time by 10%. Tech: **TensorFlow, Docker, Flask**
- Built a health analyzer application using **unsupervised learning** to predict tool failures to aid scientists and engineers. Tech: **Python, TensorFlow, FastAPI**

### CO-FOUNDER & SENIOR TECHNICAL ADVISOR | OPTIMALEX

Remote | 2021 - Present

- Architected the core NLP pipeline that analyzes public filings, deal terms, and market signals to **forecast M&A deal value and litigation risk**; delivered a Dockerized micro-service (**FastAPI, PyTorch, spaCy**) with < 15ms REST latency.
- Drove product-market fit: ran model explainability demos for VC and Fortune 500 legal teams, secured seed funding and onboarded the first three paying enterprise customers.

### PROCEEDINGS CHAIR | SCIPY CONFERENCE

Austin, TX | Tacoma, WA | 2023 - Present

- Directed the **entire editorial pipeline** for three consecutive SciPy Proceedings volumes-issued CFPs, onboarded and managed **150+ reviewers**, and executed double reviews of **30+ submissions** per year, delivering **60+ peer-reviewed papers** and introduced a GitHub Actions build system that cuts production turnaround by 40%.

- Designed and led **author & reviewer workshops** (rubric-based scoring, reproducibility checks) that drove **95% review completion** and boosted proceedings impact (h-index + 15%); partnered with IEEE, Zenodo and CurveNote to secure DOI assignment, open-access release, and indexing for accepted articles.

**GRADUATE RESEARCHER | COLUMBIA UNIVERSITY, COLUMBIA LAW SCHOOL**

**New York, NY | 2018**

- Built an attention-based encoder-decoder recommender (YouChoose CNARM), and co-developed NLP models that now power OptimaLex's contractual damage prediction engine.

CLIENT IMPACT AND CONSULTING EXPERIENCE

- Managed 15+ client engagements across Americas, the Middle East and Asia, delivering AI solutions that generated \$20M+ in business value
- Led technical presentations for C-level executives, translating complex ML concepts into strategic business insights
- Collaborated with multidisciplinary teams of domain experts, engineers, and business analysts to deliver solutions
- Experience working across diverse industries and international markets, with proven ability to adapt solutions to unique client environments

PATENTS

- IS23.1285 Wellsite Operations Machine Vision Framework
- IS23.1230 SRP Operations Diagnosis Solutions
- US-18/891,485 Artificial Intelligence-Driven Real-Time Dynamometer Classification for Diagnosis of SRPs
- US-63/300,121 Machine Vision Workflow for Wellsite Operations
- US-63/300,122 Real-Time ESP Smart Alarms Suite Enabled Through Data Analytics and Edge-Based Multiphase Flow Simulator
- US-63/358,189 Failure Prediction for Run-Life Estimation of ESPs for PHM
- US-63/389,627 Machine Learning Model Operationalization Management Framework for Continuous Updates on IIoT Devices

PUBLICATIONS

- SciPy Proceedings 2025 An Exemplar for Publishing Computational Open Science
- 2024 ADIPEC Enhancing Edge-Based SRP Production Optimization Algorithm with Fast Loop Mitigation
- 2024 SPE CETC Real-Time Well Constraint Detection Using an Intelligent Surveillance System
- 2022 ATCE Real-time Smart Alarms Suite Enabled Through Data Analytics and Edge-based Virtual Flowmeter
- 2021 SPE Symposium Electrical Submersible Pump Prognostics and Health Monitoring using Machine Learning and NLP

SKILLS

|                                |                                                                                                          |
|--------------------------------|----------------------------------------------------------------------------------------------------------|
| AI/ML Frameworks & Tools ..... | TensorFlow, PyTorch, Keras, Sklearn, OpenCV, MLFlow                                                      |
| Generative AI & LLMs .....     | Azure Foundry, Cognite Data Foundation, Dataiku, OpenAI API, LangChain, RAG implementations, Fine-tuning |
| Cloud Platforms .....          | Azure ML, Google Vertex AI, Dataiku                                                                      |
| Data Engineering .....         | Python, SQL, Pandas, Spark, Docker                                                                       |
| DevOps & MLOps .....           | Git, Azure DevOps, CI//CD, Dataiku, FastAPI, Docker                                                      |
| Programming Languages .....    | Python, SQL, R, JavaScript, C++                                                                          |

AWARDS

|                                            |                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------|
| Performed by SLB Award 2022 .....          | for Technical Excellence and Achievements in Cross-Domain AI/ML Product Development             |
| MLOps World Winner 2021 .....              | Dataiku Box of Awesome Challenge                                                                |
| Performed by SLB Award 2021 .....          | for Exceptional Development and Management of a High-Priority International Stakeholder Project |
| Petro.AI Hackathon Silver Award 2020 ..... | for Harris County Public Health Hacking for Houston                                             |
| SLB BeOutstanding Award 2020 .....         | for Excellence in Client Engagement and Satisfaction through Building AI and ML Solutions       |